**MSc Artificial Intelligence**

**Deep Learning and Generative AI**

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## Introduction

Informal settlements continue to be a major problem in many low- and middle-income regions rapid urbanization occurs much faster than it does in terms of planned development and related infrastructure. The need to identify the locations of slums accurately is critical for local governments, international humanitarian agencies, planners of public works, and those involved in sustainable development efforts. Mapping slums is a laborious process and generally conducted by human mappers who conduct assessments over large geographic areas. Because of this, remote sensing technologies (satellite imagery) and deep learning techniques have been gaining popularity as methods for automatically detecting slums. These two technologies offer advantages in terms of being able to cover larger geographic areas more quickly than human mappers.

Recently, studies have demonstrated that deep learning networks can extract relevant features in remote sensing data including: density, textures, arrangements of buildings, and surface properties associated with informal settlements. Specifically, among recent developments in deep learning networks, the Mixture-of-Experts (MoE) architecture holds promise for identifying slums because each MoE sub-network can focus on extracting specific feature patterns which a common feature extractor will capture more general representations. The selected paper, Generalizable Slum Detection from Satellite Imagery with Mixture-of-Experts, proposes such a framework to address the challenge of domain shift in slum detection across regions. In Part 1, this paper was critically appraised and extended through a proposal for an uncertainty-aware Mixture-of-Experts model using Monte Carlo Dropout.

In part 2 of this project, a working deep learning artifact for detecting slums via satellite images using PyTorch will be implemented. To account for limitations in terms of data availability and resource constraints within the project's scope, the current project will utilize satellite data only over Mumbai instead of adopting a full adaptation approach applicable across multiple regions. Three different versions of the artefact will be evaluated in this project; a baseline version that utilizes a Convolutional Neural Network (CNN), a version utilizing an Mixture-of-Experts (MoE) model and a version utilizing both an MoE model and Monte Carlo dropout at inference.

The aim is to evaluate whether the expert-based architecture and uncertainty estimation can improve slum classification performance and provide more reliable predictions in ambiguous cases.

## Background and Related Work

Automated detection of slums via satellite images has gained attention among researchers due to the costliness, difficulty of updating, and time required for traditional field surveys. According to a recent meta-analysis of slum mapping studies conducted since 2014 and up until 2024, the combination of remote sensing and deep learning is currently being utilized within this area, however the meta-analysis indicated that no universal method exists to create a generalized model that will capture all variations of slum characteristics across geographically disparate areas. Therefore, generalizability is one of the primary challenges facing researchers in this area [1].

In addition to addressing the issue of generalizability, another significant trend in remote sensing is the development of convolutional neural networks (CNNs). The reason why CNNs are so successful at extracting hierarchical spatial features (i.e., edges, texture, local shape, neighborhood pattern) is because they do so from raw image data. In fact, a comprehensive review of deep learning for remote sensing image classification reported that CNN-based methodologies are now considered to be common place for both scene-level and pixel-level classification tasks [2]. This may help explain why many researchers utilize CNNs as a strong baseline prior to evaluating other potentially more complex architectures.

Researchers have also attempted to apply transfer learning to address issues associated with slum mapping. For example, Verma et al. demonstrated that transfer learning can aid in the identification of urban slums using pre-trained convolutional networks along with high-resolution satellite imagery [3]. This is particularly important because remote sensing projects typically face challenges regarding the availability of labeled ground truth data, especially in resource-poor settings.

Another line of research addresses the application of segmentation models versus patch-level classification. Architectures such as U-Net are specifically influential due to the ability of U-Nets to retain fine-grained spatial detail while also providing sufficient contextual information in order to accurately perform pixel-level region boundary delineation [4]. Although U-Net was originally developed for biomedical image segmentation, it has been extensively adopted as a generic segmentation architecture, including those applied in remote sensing workflows, where accurate spatial boundaries are critical.

This selected paper expands upon existing research by introducing GRAM (Generalized Region-Aware Mixture-of-Experts) for detecting slums. Unlike previous methods that rely upon a single decision pathway, this paper utilizes a Mixture-of-Experts (MoE) design wherein multiple expert sub-networks specialize in identifying various types of feature patterns and the output of each sub-network is combined by a gating mechanism. The authors contend that this MoE design is beneficial because slum morphologies vary significantly across cities and regions, thus resulting in a considerable degree of domain shift [5]. As such, the authors combine expert specialization with adaptation mechanisms to improve performance on unseen locations and report results on a large-scale dataset that spans twelve cities across four continents.

Additionally, recent research has emphasized estimating uncertainty. Since standard neural networks can produce erroneous classifications with high confidence levels, this is detrimental in socially sensitive applications such as slum mapping. Gal and Ghahramani showed that dropout could be interpreted as approximating Bayesian inference and that Monte Carlo Dropout could be applied at test-time to estimate predictive uncertainty through repeated stochastic forward passes. This represents a practical method to evaluate the reliability of a prediction without employing a fully Bayesian neural network [6]. As such, this is advantageous for high-stake urban mapping tasks, as it allows researchers to differentiate certain classifications made with confidence from more ambiguous classifications.

Collectively, the literature illustrates that researchers have employed three primary approaches to address slum mapping and similar remote sensing problems: CNN-based classification, transfer-learning and segmentation models for spatial mapping and more recently expert-based or uncertainty aware methodologies to account for heterogeneity and predictability. Furthermore, the existing research establishes the basis for this study's utilization of a CNN baseline model, a Mixture-of-Expert model and an uncertainty-aware extension model.

## Methodology

This project implemented a practical deep learning pipeline for patch-based slum classification using satellite imagery. The project was developed in Python with PyTorch and three model setups were implemented: a base-line Convolutional Neural Network (CNN), a Mixture-of-Expert (MoE) model and uncertainty-aware extension using Monte Carlo Dropout. Even though the paper's authors suggested the use of their GRAM framework for detecting slums across regions, the implementation data that is available for this study is limited to one city: Mumbai. Therefore, the methodology provides a simplified, yet possible, implementation of some of the architectural concepts presented in the paper, but does not provide a complete reproduction of the GRAM framework.

### Dataset and Patch Preparation

NumPy array datasets are loaded into the program which include a 4-channel satellite image of shape (4, 3954, 2105) and a corresponding slum label mask of shape (3954, 2105). Three of the four channels were utilized for RGB visualization purposes, while all four channels were utilized for training purposes. An overlay of the RGB image, the label mask, and the labeled area visually confirms that the label mask matches the area shown in the RGB image.

Since the entire image was a large single scene, a sliding window method was employed to divide the images into smaller patches. The size of each patch was 64 x 64 pixels and the step-size of the sliding window was 64 pixels. This produced non-overlapping patches of images. For each patch, the corresponding portion of the label mask was retrieved and converted into a binary classification. If the average value of the mask was above 0.01 for a particular patch, then the patch was classified as part of a slum; otherwise it was classified as non-slum. This resulted in 1,952 patches being created with dimensions (1952, 4, 64, 64). While this patch-level approach greatly reduced computational complexity for classification tasks, it did simplify the original mapping task to a binary classification problem.

### Data Splitting and Preprocessing

The dataset was split into training, validation and test sets using stratified splitting to maintain class balance. In total, Seventy percent of the data were assigned to training, while the remaining thirty percent were divided equally between validation and testing. Thus, there were 1,366 training patches, 293 validation patches and 293 test patches. Channel-wise statistics were

calculated from the training set only to normalize the validation and test sets. Following this, a custom PyTorch Dataset class and two DataLoader objects were created for batched processing. Because there was some degree of imbalance in classes in this dataset, class weights were computed and included in the cross-entropy loss function so that the minority slum-class would have increased weighting when being trained.

**Model Architectures**

A baseline CNN was implemented as the reference model. It comprised of three convolutional blocks where each block contained: convolution layer; batch normalization layer; ReLU activation layer; and max-pooling layer. These convolutional blocks were followed by several fully-connected layers with dropout. This architecture served as a simple, yet effective, benchmark for patch classification. The second model was a Mixture-of-Experts architecture with the same shared convolutional feature-extractor as the previous model; three expert-sub-networks; and a gating-network that would output soft-weights to combine expert-output. This allowed the model to allocate representational capacity more selectively than the baseline CNN.

**Uncertainty-Aware Extension**

To align the implementation more closely with the Part 1 proposal, an uncertainty-aware extension was added to the trained MoE model using Monte Carlo Dropout. During inference time, the dropout layers within the expert networks were turned back on and multiple random passes were made over each test sample. The mean class-probability as final prediction and utilized the variance across passes that is the measure of prediction uncertainty. This provided with a practical way to evaluate prediction confidence without having to build out a full Bayesian model.

**Training and Evaluation**

Both models were trained using an Adam optimizer with a base learning rate of 0.001, batch size = 16, and maximum number of epochs = 25. Both models were stopped early if they failed to improve over five consecutive epochs based on their validation loss. Performance for both models was measured using metrics such as accuracy, precision, recall, F1-score, classification reports, confusion matrixes, and uncertainty measures for distinguishing between correct and incorrect predictions.

## Experiments and Results

### Experimental Setup

The models were trained and evaluated on stratified training, validation and test splits derived from the patch dataset. Training was performed using the Adam optimizer for up to 25 epochs with early stopping based on validation loss.

### Baseline CNN Results

The baseline CNN achieved an accuracy of 0.8089, precision of 0.6299, recall of 0.8989, and F1-score of 0.7407 on the test set.

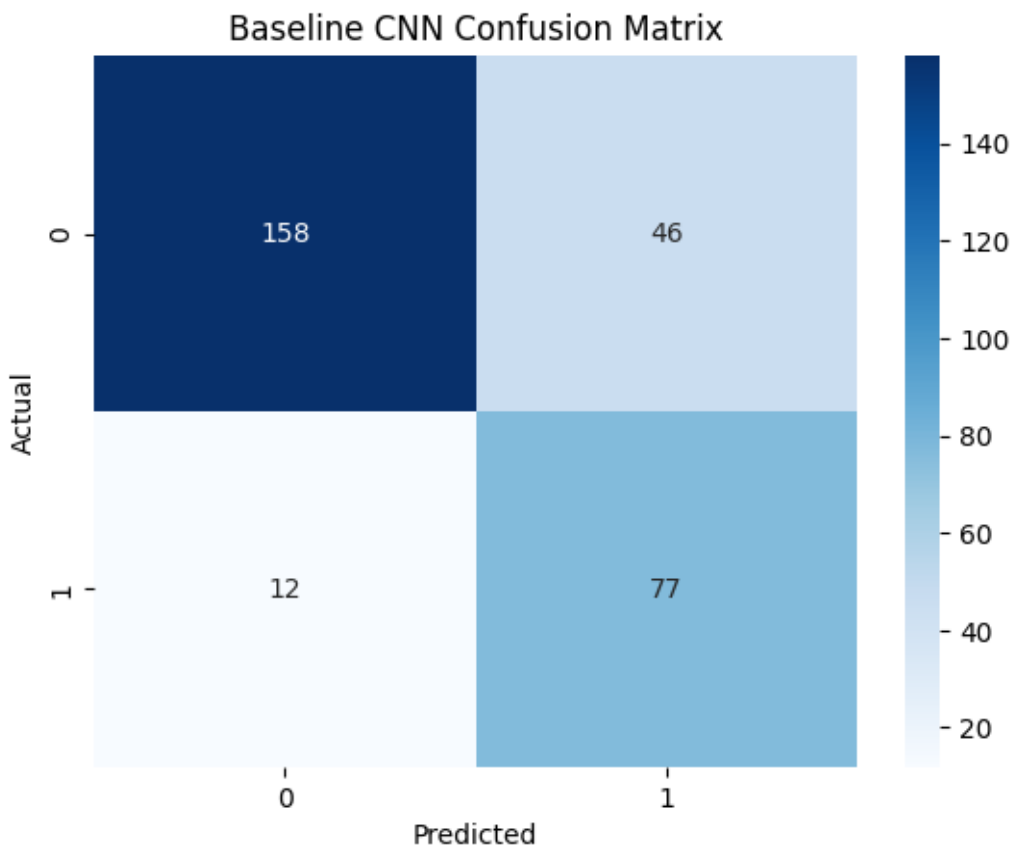


***Figure 1. Baseline CNN Confusion Matrix***

These results indicate that the baseline model was effective at identifying slum patches and thus having a high Recall value; however, the model has a low Precision metric which means many Non-Slums were mis-classified as Slums.

**Mixture-of-Experts Results**

The standard Mixture-of-Experts model achieved an accuracy of 0.7986, precision of 0.6103, recall of 0.9326, and F1-score of 0.7378.

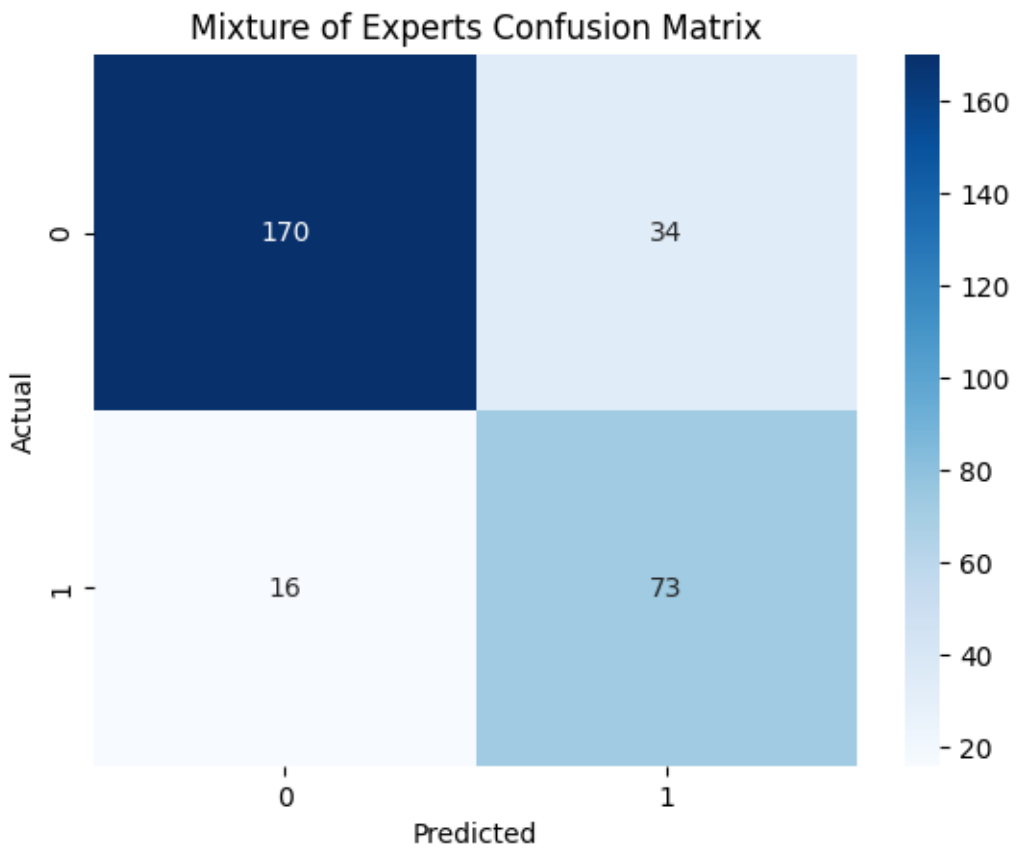


***Figure 2. Mixture-of-Experts Confusion Matrix***

In comparison to the baseline CNN, the Standard Mixture-of-Experts model found slightly more Slums than the baseline; however, the Standard Mixture-of-Experts model had a much lower overall Accuracy and Precision values. Thus, the model became more sensitive to slum patterns but at the cost of increased false positives.

**Uncertainty-Aware MoE Results**

Monte Carlo Dropout is used during inference to generate uncertainty estimates. The uncertainty aware MoE model's metrics for the test data were: Accuracy of 0.8259, Precision of 0.6759, Recall of 0.8202, and F1-Score of 0.7411.

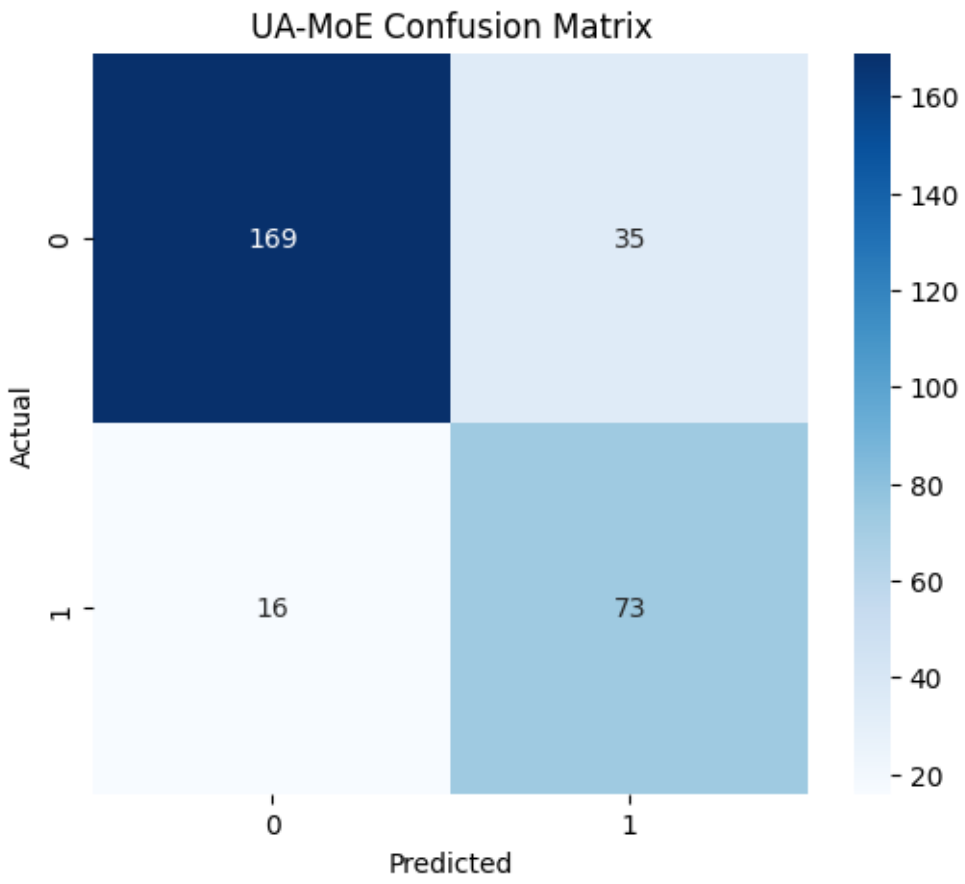


***Figure 3. UA-MoE Confusion Matrix***

This gave the best overall Accuracy out of all MoE model variations tested. Furthermore, the uncertainty aware MoE model also achieved the best F1-Score compared to the other MoE model variation. Most importantly, the average uncertainty for wrong classifications (0.00393) is significantly larger than for right classifications (0.00306). This shows that the model is less confident about making mistakes which supports that uncertainty estimates can help improve prediction reliability.

| **Model** | **Accuracy** | **Precision** | **Recall** | **F1 Score** |
|---|---|---|---|---|
| Baseline CNN | 0.8089 | 0.6299 | 0.8989 | 0.7407 |
| MoE | 0.7986 | 0.6103 | 0.9326 | 0.7378 |
| UA-MoE | 0.8259 | 0.6759 | 0.8202 | 0.7411 |

***Table 1. Model Performance Comparison***

Both the baseline CNN and UA-MoE provided F1 scores close together, and Standard MoE performed slightly below overall compared to Standard MoE, even though it had the highest level of recall.

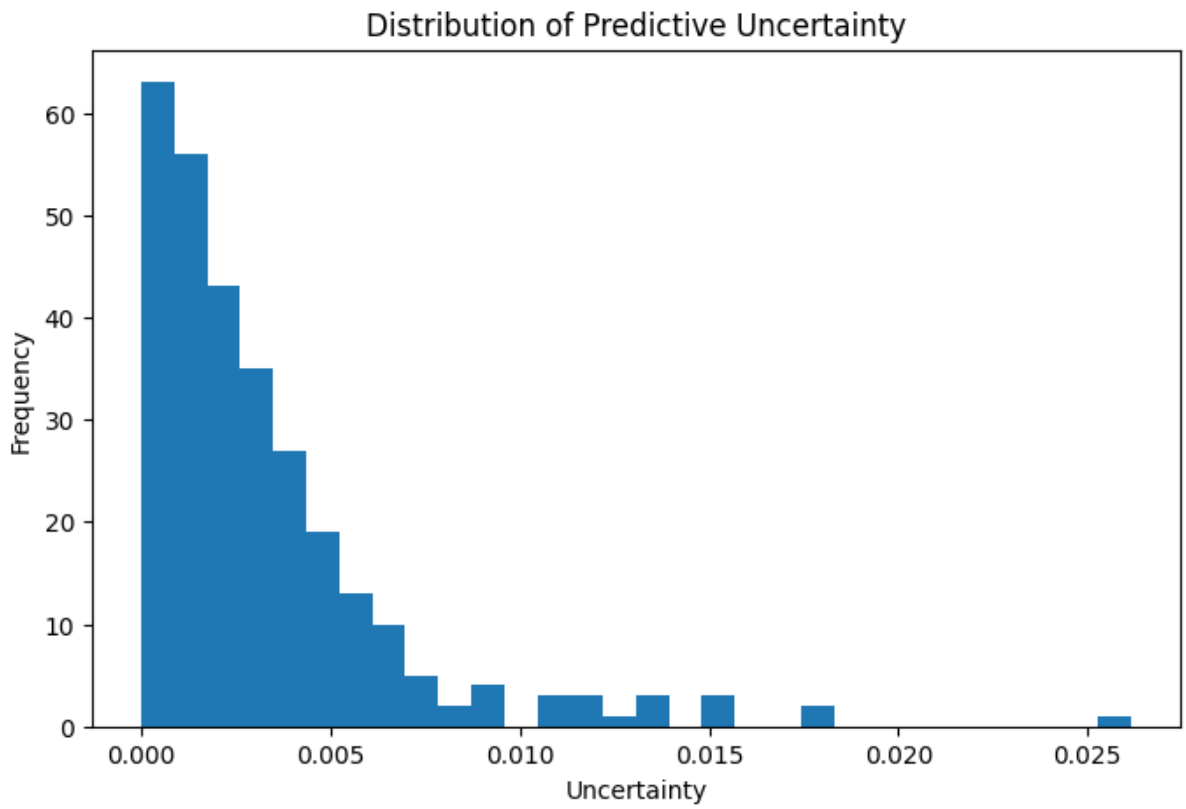


***Figure 4. Predictive Uncertainty Histogram***

However, the addition of uncertainty awareness does not significantly alter classification performance, but provides an additional reliability aspect through demonstrating that there is typically more uncertainty with incorrect classifications than those that are correct. As such, the uncertainty aware variant is generally more useful than a standard classifier, especially in situations where ambiguity exists.

## Discussion

### Interpretation of Results

The results show that patch-based slum classification from satellite imagery is feasible, although the task remains difficult because slum and non-slum areas often share similar visual characteristics. The baseline CNN achieved strong overall performance and high recall for the slum class, indicating that conventional convolutional feature extraction was effective for learning useful spatial and texture patterns from the available data.

The standard Mixture-of-Experts model increased recall, suggesting that expert specialization improved sensitivity to slum-like features. However, this did not improve overall performance, as accuracy, precision, and F1-score were slightly lower than those of the baseline CNN. This indicates a trade-off between higher sensitivity and lower specificity.

The uncertainty-aware MoE was the most meaningful result in relation to Part 1. Although it did not significantly increase F1-score, it achieved the highest accuracy among the MoE-based variants and showed that incorrect predictions had higher uncertainty than correct ones. This suggests that Monte Carlo Dropout improved the reliability and interpretability of model outputs.

### Challenges and Unexpected Outcomes

An unexpected finding was that the standard Mixture-of-Experts model failed to surpass the baseline CNN. Considering the theoretical merits of expert based learning, it might have been anticipated that superior performance could be demonstrated. One possible explanation is that the dataset consisted of only a single city; hence insufficient visual diversity existed across regions for expertise to provide significant advantages. Alternatively, the additional flexibility afforded by the Mixture-of-Expert model may have caused the model to become overly sensitive to localized ambiguity, resulting in an excessive number of false positives.

Another challenge facing this research effort has been visual overlap between classes. Urban dense patches can include features/arrangements that resemble informal settlement areas even though they have not been labeled as such. These issues create inherently difficult boundaries for classification and help account for why all models exhibited a much stronger recall rate than precision for the slum class. The uncertainty aware variant somewhat alleviated this issue

through demonstration that incorrect classifications tended to exhibit higher uncertainty values, but the difference between correct and incorrect uncertainty values were still quite small, indicating that further refinements would be necessary.

**Alignment with the Project Objectives**

The project aimed to implement and evaluate a deep learning solution inspired by the selected paper and to test whether an uncertainty-aware extension could improve prediction reliability. This objective was achieved in a reduced but meaningful form. A baseline CNN, a Mixture-of-Experts model, and an uncertainty-aware extension using Monte Carlo Dropout were successfully implemented. The results showed that incorrect predictions had higher uncertainty than correct predictions, supporting the Part 1 argument that uncertainty awareness can improve reliability and interpretability.

However, the full GRAM framework was not reproduced. The original paper focuses on cross-region generalization, pseudo-label refinement, and adaptation to unseen cities, whereas this study used a single-city dataset and supervised patch classification. As a result, the reliability objective was supported, but the cross-region component of the Part 1 hypothesis could not be tested directly.

**Limitations and Future Work**

The main limitation of the project is the dataset scope. Because only one Mumbai scene was available, cross-city or cross-region generalization could not be evaluated. In addition, the task was simplified from spatial slum mapping to binary patch classification, which reduced the spatial complexity of the original problem.

Another limitation is that uncertainty was incorporated only at inference time. More advanced proposal elements, including uncertainty-guided gating, pseudo-label filtering, and calibration metrics such as Expected Calibration Error, were not implemented. Therefore, the final model should be understood as a simplified approximation of the proposed UA-MoE framework rather than a full version.

Future work should use a multi-city dataset, integrate uncertainty directly into the gating mechanism, and extend the study to pixel-level segmentation and calibration analysis.

## Ethics and Scalability

The use of deep learning for slum detection raises important ethical concerns because the outputs may influence decisions affecting vulnerable communities. A false positive could incorrectly label a formal area as a slum, while a false negative could cause a genuinely deprived area to be overlooked in planning or resource allocation. In either case, model error may contribute to unfair treatment, stigma, or inefficient policy action. This is especially important in this project because the task concerns socially sensitive urban environments rather than a neutral image classification problem. For this reason, model predictions should not be treated as final decisions, but as decision-support outputs that require human interpretation and contextual validation.

Bias is another key concern. The implementation used data from a single Mumbai scene, which means the learned visual patterns are likely to reflect only one urban context. As a result, the model may perform less reliably when applied to other cities with different settlement structures, materials, or environmental characteristics. This limitation increases the risk of geographic bias and reinforces the importance of careful interpretation of results. The uncertainty-aware extension partly addresses this issue by indicating when the model is less confident, which can help identify predictions that may require additional review.

Scalability must also be considered. The baseline CNN is relatively efficient, whereas the Mixture-of-Experts model introduces extra computational cost through multiple expert heads. The uncertainty-aware extension adds further overhead because Monte Carlo Dropout requires repeated inference passes. While this remains manageable at the current patch-based scale, deployment over larger urban regions would require greater storage, processing time, and compute resources. Therefore, practical real-world use would require a balance between predictive reliability and computational efficiency.

## Conclusion

This project developed and evaluated a deep learning pipeline for slum classification from satellite imagery using a baseline CNN, a Mixture-of-Experts model, and an uncertainty-aware extension based on Monte Carlo Dropout. The results showed that patch-based slum detection is feasible and that a conventional CNN remains a strong baseline for this task. Although the standard MoE model did not outperform the baseline overall, the uncertainty-aware extension

added an important reliability component by showing that incorrect predictions were associated with higher uncertainty than correct predictions.

The main significance of the project is therefore not only its classification performance, but its demonstration that deep learning models for socially sensitive applications should be evaluated in terms of reliability as well as accuracy. The study also highlighted the practical challenges of aligning research-level frameworks such as GRAM with limited datasets and implementation constraints. Overall, the project shows that deep learning is a promising approach for slum detection from satellite imagery, but that uncertainty estimation and careful interpretation are essential if such systems are to support responsible real-world decision-making.

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